

Estimating disease importation risk during the 2026 FIFA World Cup

High-level summary, limitations, and paths forward

- **Importation risk is disease-specific and regionally structured.** Expected importations are dominated by different sets of origin countries depending on the disease considered, reflecting geographic patterns of endemicity and outbreaks. For example, dengue importations are primarily associated with countries in the Americas, measles and pertussis importations are dominated by European countries, and malaria importations are driven largely by African countries.
Limitation: Disease incidence and prevalence estimates rely on reported surveillance data, which vary substantially in completeness and timeliness across countries.
Potential improvement: Incorporate disease- and country-specific underreporting corrections, regional adjustment factors, or ensemble prevalence estimates derived from multiple surveillance sources.
- **International entry into the United States is highly concentrated at major gateway airports.** Approximating airport entry using passenger enplanement volumes indicates that a small number of large hubs account for a substantial fraction of international arrivals, leading to geographic clustering of importation risk at entry points.
Limitation: Airport entry probabilities are inferred from total passenger boardings, ignoring airline network structure, hub specialization, and country-specific routing patterns.
Potential improvement: Replace or augment enplanement-based probabilities with origin–destination flight data, airline schedule data, or modeled air-traffic networks linking countries to specific airports.
- **Relative importation magnitudes reflect baseline surveillance data and simplifying travel assumptions.** Under baseline assumptions—reported surveillance data accurately reflect infectious prevalence and infected individuals are equally likely to travel—differences in expected importations across diseases and origin regions are driven primarily by reported incidence and seasonal patterns.
Limitation: Underreporting and disease-specific travel behavior (e.g., symptom severity, screening, travel deterrence) are not explicitly modeled.
Potential improvement: Introduce disease-specific asymptomatic fractions, travel propensities, and reporting multipliers, with uncertainty propagation through sensitivity or probabilistic analyses.
- **Expected importations should be interpreted as intensities, not outbreak guarantees.** The estimated quantities represent expected numbers of infected arrivals and probabilities of at least one introduction, rather than predictions of sustained transmission or outbreaks within host cities.

Limitation: Importation events are assumed to be independent and Poisson-distributed, neglecting potential clustering of infected travelers on specific flights.

Potential improvement: Incorporate overdispersed arrival processes or flight-level clustering models to better capture correlated importation events.

- **Results represent a conservative baseline for subsequent analyses.** The current estimates likely understate true importation risk, as they do not account for underreporting, asymptomatic infections, heterogeneous travel behavior, or event-driven deviations from historical travel patterns.

Limitation: Travel volumes and behavior during the World Cup are approximated using historical averages rather than event-specific dynamics.

Potential improvement: Integrate tournament-specific travel surges using ticket-holder nationality data, match schedules, or scenario-based perturbations to baseline travel patterns.

1 Initial idea(s)

I am thinking of three potential analyzes that could be made pretty much independently and with different data sources.

1. **Importation risk:** transmission of infections from origin countries to U.S. host cities through international travel.
2. **Local outbreak risk:** amplification within host cities given susceptibility, vaccination, demography, and event-driven density.
3. **Domestic propagation:** spread from host cities to other U.S. regions through mobility flows.

1.1 Importation risk – data requirements

1. Estimated prevalence of diseases in each origin country (Dengue, Malaria, Measles, Influenza, COVID-19, and others), preferably at monthly or weekly resolution to capture seasonal patterns. When monthly data are not available, we interpolate prevalence assuming smooth seasonal trends or using historical averages.
2. International passenger flows (number of incoming travelers) by country of origin. If daily data are not available, we approximate the distribution of arrivals around match dates using proportional allocation based on (a) ticket-holder nationality, (b) historical tourism inflow patterns, and (c) airline seat capacity.
3. Nationality distribution of FIFA ticket holders, used to scale baseline travel volumes and construct tournament-specific travel surges.
4. Airport-of-entry data to map international arrivals to specific host cities. When travelers transit through multiple airports, we distribute arrivals proportionally to historical domestic flight connections.
5. Disease-specific parameters: underreporting factors $\rho_{c,d}$, infectious periods L_d , asymptomatic fractions a_d , and symptomatic travel probabilities s_d . When country-specific values are unavailable, regional or global averages could be used, and uncertainty ranges are incorporated into sensitivity analyses.

1.2 Importation risk – modeling framework

Let c denote origin country, h host city, and d disease. Let $T_{c \rightarrow h, t}$ denote traveler volume from c to h at time t .

Given incidence $I_{c, d, t}$ (new cases at time t), population N_c , underreporting factor $\rho_{c, d}$, and infectious period L_d , estimated proportion of the population that is currently infectious is:

$$\pi_{c, d, t} = \frac{\rho_{c, d} \sum_{\tau=t-L_d}^t I_{c, d, \tau}}{N_c}.$$

Expected number of infectious arrivals from c to h :

$$\lambda_{c \rightarrow h, d, t} = T_{c \rightarrow h, t} \times \pi_{c, d, t} \times p_{\text{travel}|\text{infectious}, d}.$$

The travel-behavior parameter is decomposed as:

$$p_{\text{travel}|\text{infectious}, d} = a_d + (1 - a_d)s_d,$$

where a_d is the asymptomatic fraction for disease d and s_d the probability that a symptomatic infected individual completes travel.

Total importation intensity (expected number of imported infected individuals) into host city h over the tournament window $\mathcal{T}$:

$$\Lambda_{h, d}^{\text{import}} = \sum_c \sum_{t \in \mathcal{T}} \lambda_{c \rightarrow h, d, t}.$$

Assuming introductions follow a Poisson process, the probability of at least one infectious arrival is:

$$\mathbb{P}(X_{h, d} \geq 1) = 1 - e^{-\Lambda_{h, d}^{\text{import}}}.$$

Note. On the side of travels, the data I have access to are:

- Monthly international visitor arrivals by country of origin (COR) from 2000 to September 2025 [1].
- Ranking of airports by volume of enplaned passengers in 2023 [2].

So, I thought of an approximation due to the lack of more accurate data. If

$$T_{c \rightarrow USA, t} = \text{traveler volume from country } c \text{ to any airport in the USA.}$$

Then, assuming that there is a strong and positive correlation between enplaned and deplaned passengers in an airport, we can approximate the “probability of arrival to an airport” as proportional to the volume of “enplanement” of airport a as

$$\text{Prob. of arriving to airport } a = \frac{V_a}{\sum_a V_a}.$$

Finally, assuming that arrivals to the USA follow this probabilities

$$T_{c \rightarrow h, t} \approx T_{c \rightarrow USA, t} \times \text{Prob. of arriving to airport } a$$

1.2.1 Analysis with available data: Importation risk

This section describes the data sources and assumptions used to estimate the risk of disease importation into the United States during the FIFA World Cup 2026 period. A key limitation of the available data is the absence of information on direct passenger flows from country c to specific U.S. airports h . As a result, importation risk is estimated using a combination of country-level arrival volumes and airport-level entry probabilities.

1. **International arrivals by country.** I use data from the International Visitor Arrivals Program (ADIS I-94), managed by the National Travel and Tourism Office (NTTO) in cooperation with the Department of Homeland Security (DHS) and U.S. Customs and Border Protection (CBP). The ADIS I-94 dataset records international visitor arrivals to the United States for stays of at least one night under eligible visa categories and is widely used to quantify travel and tourism volumes [1].

Using these data, I estimate the number of travelers arriving in the United States from each country during June–August, corresponding to peak summer travel and overlapping with the World Cup period.

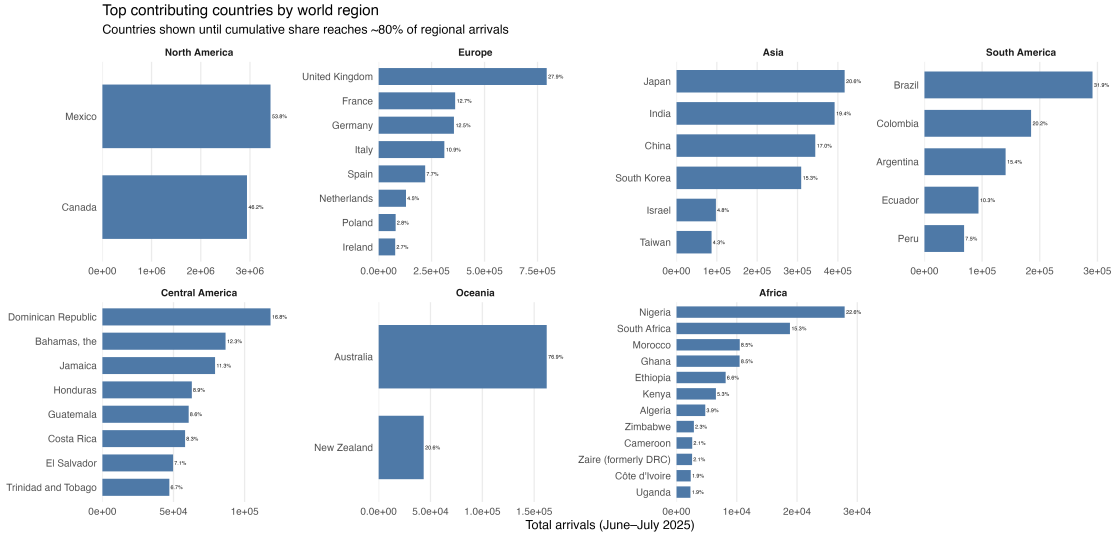


Figure 1: Distribution of international arrivals to the United States during June–July 2025 by country of origin and world region. For each region, countries are shown until their cumulative contribution reaches approximately 80% of total regional arrivals, highlighting the concentration of travel volume among a limited number of origin countries.

2. **Airport-level entry probabilities.** To approximate how international arrivals are distributed across U.S. airports, I use Passenger Boarding (Enplanement) data from the FAA Air Carrier Activity database, compiled by the Bureau of Transportation Statistics (BTS). These data report total passenger boardings for each airport and are used for airport classification and funding allocation under the Airport Improvement Program (AIP) [2].

Assuming that international arrivals are approximately proportional to total passenger boardings, the probability that an arriving international traveler enters the United States through

airport a is estimated as

$$\text{Prob. of arriving at airport } a = \frac{V_a}{\sum_a V_a},$$

where V_a denotes the total number of passenger boardings at airport a .

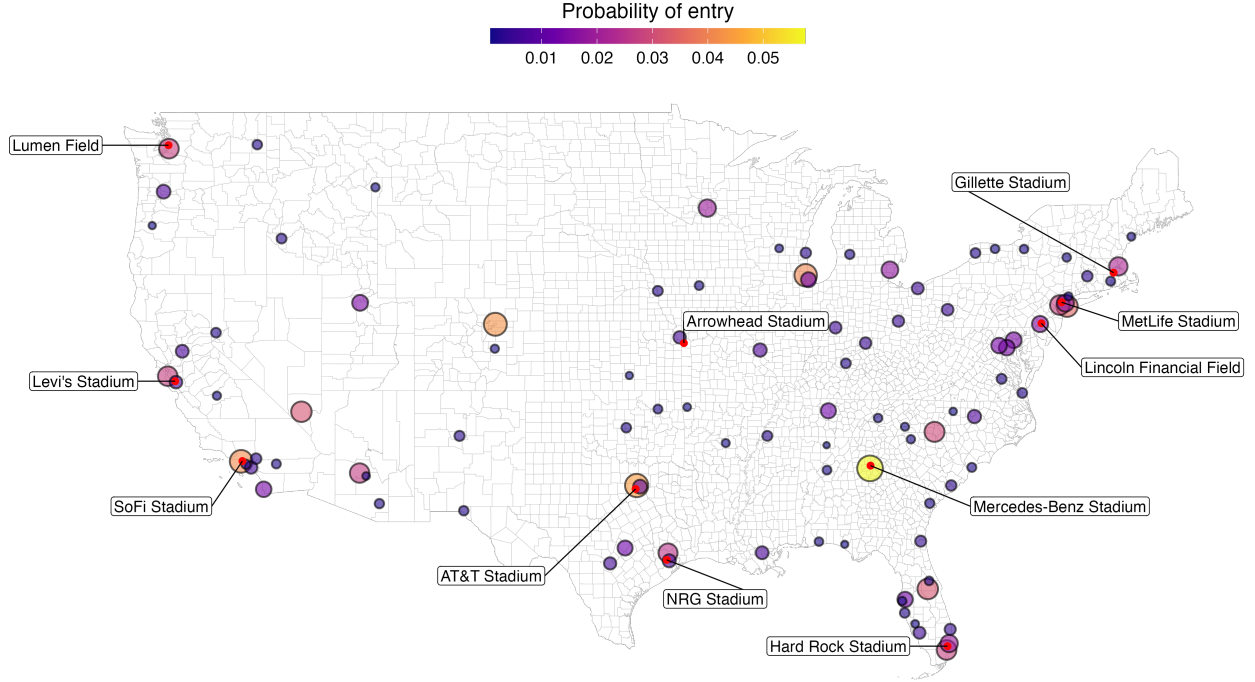


Figure 2: Spatial distribution of airport entry probabilities across the contiguous United States. Circles represent airports, with size and color proportional to the estimated probability of international entry. Locations of selected FIFA World Cup 2026 venues are shown for geographic reference.

3. **Approximating country-to-airport flows.** Combining country-level arrival volumes with airport-level entry probabilities, I approximate the number of travelers arriving from country c to airport h at time t as

$$T_{c \rightarrow h, t} \approx T_{c \rightarrow USA, t} \times \text{Prob. of arriving at airport } h.$$

This approximation assumes that the spatial distribution of international arrivals across U.S. airports is independent of country of origin. While this assumption neglects airline routing structure and regional connectivity, it provides a tractable first-order estimate in the absence of origin–destination airport data.

1.2.2 Disease-specific importations

Given the estimated travel volumes, I next estimate disease-specific importation risk. Airport-level entry probabilities are assumed to be independent of disease status; therefore, differences in importation risk across diseases arise solely from differences in disease incidence or prevalence in origin countries.

For disease d , the expected number of infected travelers arriving in the United States from country c at time t is

$$\lambda_{c \rightarrow USA,d,t} = T_{c \rightarrow USA,t} \times \pi_{c,d,t},$$

where $\pi_{c,d,t}$ denotes the estimated proportion of the population in country c that is infectious at time t .

At this stage, the underreporting factor $\rho_{c,d}$ and the probability of traveling while infectious $p_{\text{travel}|\text{infectious},d}$ are fixed to unity. This implies that reported surveillance data accurately reflect infectious prevalence and that infectious individuals are equally likely to travel. These simplifying assumptions are used to obtain baseline estimates and can be relaxed in future sensitivity analyses.

The total importation intensity for disease d over the study period $\mathcal{T}$ is

$$\Lambda_{USA,d}^{\text{import}} = \sum_c \sum_{t \in \mathcal{T}} \lambda_{c \rightarrow USA,d,t}. \quad (1)$$

Assuming importation events follow a Poisson process, the probability of observing at least one infectious arrival in the United States during the study period is

$$\mathbb{P}(X_{USA,d} \geq 1) = 1 - e^{-\Lambda_{USA,d}^{\text{import}}}. \quad (2)$$

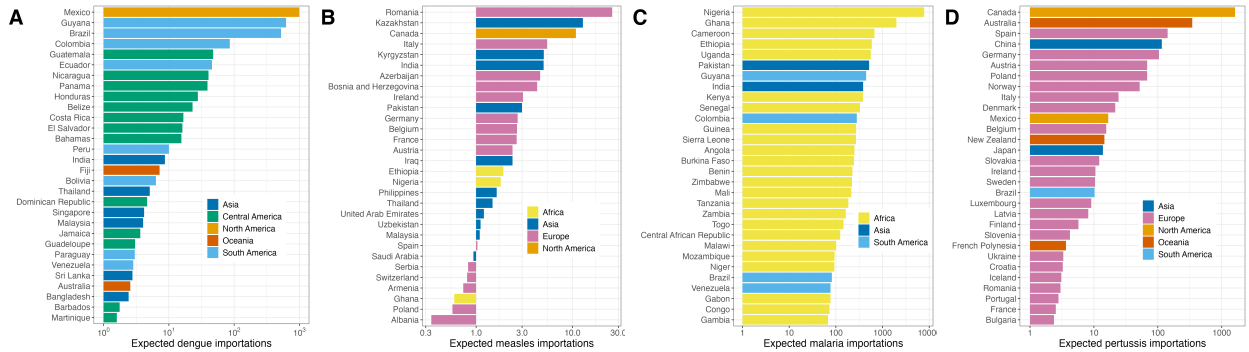


Figure 3: Top contributing countries for expected importation of selected infectious diseases into the United States during the World Cup period. Panels show the 30 countries with the highest expected importations for (A) dengue, (B) measles, (C) malaria, and (D) pertussis. Expected values are estimated at the country level by combining disease surveillance data with international travel volumes. Values correspond to expected counts for the entire United States, prior to allocation to specific airports or host cities.

1.2.3 Limitations and potential improvements

The importation analysis presented here is subject to several limitations:

- International arrival data are aggregated at the national level, preventing direct estimation of country-to-airport passenger flows.

- Airport entry probabilities are inferred from total passenger boardings, ignoring airline network structure, hub specialization, and country-specific routing patterns.
- Disease incidence and prevalence estimates rely on reported surveillance data, which may vary substantially in completeness and timeliness across countries.
- Underreporting and disease-specific travel behavior are not explicitly modeled, potentially biasing relative importation risks.
- Importation events are assumed to be independent and Poisson-distributed, neglecting possible clustering of infected travelers on specific flights.
- Temporal variation in travel behavior during large international events may differ from historical averages used in this analysis.

References

- [1] U.S. Department of Commerce, International Trade Administration. *I-94 Arrivals Program*. Available at: <https://www.trade.gov/i-94-arrivals-program> (Accessed: January 2, 2026).
- [2] Bureau of Transportation Statistics (BTS). *Airport Rankings 2023*. Available at: <https://www.bts.gov/topics/airlines-and-airports/airport-rankings-2023> (Accessed: January 2, 2026).
- [3] Pan American Health Organization. Dengue: Data, Maps and Statistics. Available at: <https://www.paho.org>.
- [4] World Health Organization. Measles surveillance data. Available at: <https://www.who.int>.